



CMIP6 model evaluation for sea surface height responses to ENSO

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Abstract

El Niño–Southern Oscillation (ENSO) exhibits a substantial impact on sea surface height (SSH), and some climate models are significantly biased in reproducing the observed ENSO impact on SSH. This study aims to assess the performance of 26 Coupled Model Intercomparison Project Phase 6 (CMIP6) models for global SSH response to ENSO using their historical simulations from 1958 to 2014 during December–February (DJF). The observation data for SSH and sea surface temperature (SST) are derived from Ocean Reanalysis System 5 (ORAS5). The fidelity of models is evaluated based on skill metric parameters such as root mean square error, interannual variability skill score, and M-Score. In CMIP6 models, El Niño plays a dominant role to simulate enhanced SSH across eastern Pacific Ocean (Region I) and southwest tropical Indian Ocean (Region III). In contrast, El Niño induces cooler SSTs over the region bounded between 100°E–133°E, 5°S–20°N (Region II), which is further associated with reduced SSH. The inter-model relationship of ENSO amplitude with the area mean regression coefficient for SSH and SST is investigated for these three hotspot regions. Overall, models with stronger ENSO amplitude simulate higher values of SSH and SST over Region I, III, and weaker amplitude over Region II.

Keywords CMIP6 · Sea surface height · Global Ocean · ENSO · Model evaluation

1 Introduction

Sea surface height (SSH) is a crucial oceanographic parameter (Combes and Matano 2019), and provides a greater understanding of various aspects of ocean dynamics, such as the ocean's response to large-scale forcing, wave propagation, and heat storage (Donohue et al. 2016). These factors are important in forecasting and mitigating the effects of climate change, as the ocean plays a pivotal role in regulating the Earth's climate. SSH is also an important indicator of sea level change, and scientists are continuously monitoring it due to its significant impacts on coastal communities, infrastructure, and ecosystems (Kumar et al. 2021; Li et al. 2022; Magnan et al. 2022; Chen et al. 2023). Moreover, the recent acceleration in global SSH from 1993 to 2019 is

anticipated to have potentially catastrophic regional effects (Barrier et al. 2013; Prandi et al. 2021), further highlighting the importance of understanding SSH to prevent the associated social and economic disasters.

Previous studies have analyzed the trends in SSH over the north Atlantic Ocean (AO) (Häkkinen et al. 2013; Brunabend et al. 2014), north Pacific Ocean (PO) (Schloesser et al. 2021), Arctic region (Armitage et al. 2016), Indian Ocean (IO) (Salim et al. 2012; Zhang and Mochizuki 2022), and Bay of Bengal (Ghosh et al. 2017). Over the AO, studies have found a significant influence of the North Atlantic Oscillation on SSH (Wang et al. 2013). The Kuroshio Extension (KE), lying along the western side of the north PO, exhibits large SSH variability (Joh et al. 2022). Studies report that the basin-wide wind changes over the midlatitudes are the primary forcing of the decadal variability of SSH fluctuations in KE (Qiu and Chen 2010; Tamura and Tozuka 2023). The SSH variability in the tropical PO can also propagate into the south IO through the Indonesian Archipelago and along the west coast of Australia (Menezes and Vianna 2019). Additionally, Swapna et al. (2017) analyzed various components of SSH over the IO and found it

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to be dominated by the steric component, with further dominance by the thermosteric component.

Importantly, variations in SSH indicate the ocean's heat content (Li et al. 2022). Oceans absorb around 90% of the heat supplied to the global climate systems (Johnson and Lyman 2020; von Schuckmann et al. 2023). The spatial distributions of SSH, surface wind stress, and SST are closely associated, especially in the tropics (Flato et al. 2013; Yang et al. 2022; Dhage and Widlansky 2022). Amplification of SST anomalies over the tropical PO results in El Niño–Southern Oscillation (ENSO) phenomenon through the positive ocean–atmosphere Bjerknes feedback (Bjerknes 1969). Planton et al. (2021) describe the wind–stress–SSH feedback that links the remote trade wind anomalies to the subsurface temperature anomalies in the eastern equatorial PO, and the SSH–SST feedback links the subsurface temperature anomalies to the eastern PO SST anomalies. Variations in SSH exert a crucial role in the climate system by affecting the fluctuations in ocean currents and thermocline depth, thereby significantly impacting the model simulations of climate variability, such as ENSO. According to Li et al. (2022), thermocline anomalies (SSH anomalies) in the western (eastern) tropical PO positively (negatively) influence the discharge phase of ENSO, primarily due to the asymmetrical wind stress anomalies associated with El Niño and La Niña.

Global climate models (GCMs) are increasingly being utilized to understand the past and future changes in climate events, and further decision-making related to climate impact (Watterson 2019; Bi et al. 2020; Lyu et al. 2020; Raju and Kumar 2020). Model evaluation has demonstrated systematic improvement in the representation of modes of variability across the CMIP archives, particularly for ENSO and the Pacific Decadal Oscillation (Fasullo et al. 2020). Specifically among global climate models, CESM2 is highly skilled in simulating the present-day climate, especially for ENSO (Fasullo 2020). Further, a significant improvement in the time-mean SSH representation from CMIP3 to CMIP5 has been reported (Landerer et al. 2014; Lyu et al. 2015). Yet, the discrepancies between CMIP5 and CMIP6 models are less significant (Eyring et al. 2016). SSH fields in the equatorial regions of IO and PO are found to be consistent with biases in the zonal surface wind stress fields (Landerer et al. 2014). The SSH trends are expected to intensify in the future emission scenarios (RCP4.5 and RCP8.5), as the atmosphere continues to accumulate higher concentrations of greenhouse gases (Giarolla et al. 2020). Several studies are being undertaken based on the climatology and variability of SSH, and trends in SSH using CMIP5 and CMIP6 models. However, CMIP6 model evaluation based on the SSH association to ENSO is still not explored

in the literature. Therefore, it is essential to study the CMIP6 model performance using different skill parameters.

This study examines the performance of 26 CMIP6 models to reproduce the observed ENSO teleconnections with SSH globally from 1958–2014 during boreal winters. Simple linear regression analysis is utilized to obtain each model's SSH response to ENSO and compare the results with the observation data. Furthermore, an evaluation based on the skill metric parameters is conducted. For the three selected hotspot regions that display robust SSH response to ENSO, this study delves into the models' proficiency concerning the factors that are liable for simulating the regional SSH response to ENSO. In addition, the role and relative significance of the ENSO amplitude along with SST and its teleconnection with SSH patterns is also examined over the hotspot regions.

Overall, the paper is structured as follows. Section 2 elaborates on the data and methodology used in this study. Section 3 contains the climatology, variability, and findings of the CMIP6 model assessment for the global ocean and the three hotspot regions. Section 4 presents the discussion and conclusion.

2 Data and methodology

2.1 Data

Table 1 presents 26 CMIP6 historical models employed in this study to assess their performance. The analysis of SSH and SST over the global ocean is based on the models' output during December–February (DJF). Access to the CMIP data is facilitated by the Earth System Grid Federation (ESGF), one of the most intricate big data systems (Williams et al. 2015; Schuckmann et al., 2023). The CMIP6 archive, available at the URL <https://esgf-node.llnl.gov/projects/cmip6/>, provided the monthly historical model data for the period 1958–2014. The 'zos' and 'tos' variables from the CMIP6 output group OMON (for ocean data) correspond to SSH and SST, respectively. The study considered data from the first ensemble member-run of variant label "r1i1p1f1" for CMIP6 experiments. Here, r1 is the realization index, i1 initialization index, p1 physics index, and f1 forcing index (Taylor et al. 2018). The CMIP6 models exhibit different horizontal resolutions. Thus, the models were interpolated to a common $1^\circ \times 1^\circ$ grid using bilinear interpolation to enable easier comparison (See Table 1 for original resolution).

Further, SSH and SST data obtained from the Ocean Reanalysis System 5 (ORAS5) generated by the European Centre for Medium-Range Weather Forecasts (ECMWF) is used as an observation data to compare with the CMIP6 models (Zuo et al. 2019). This reanalysis product

Table 1 26 CMIP6 models used in this study

Model Number	Model Name	Country/Region	Resolution (lon by lat)	Ocean Component	Atmosphere Component
1.	ACCESS-CM2	Australia	360 × 300	ACCESS-OM2	MetUM-HadGEM3-GA7.1
2.	ACCESS-ESM1-5	Australia	360 × 300	ACCESS-OM2	HadGAM2
3.	BCC-CSM2-MR	China	360 × 232	MOM4	BCC_AGCM3_MR
4.	CAMS-CSM1-0	China	360 × 200	MOM4	ECHAM5_CAMS
5.	CanESM5	Canada	361 × 290	NEMO3.4.1	CanAM5
6.	CAS-ESM2-0	China	362 × 196	LICOM2.0	IAP AGCM 5.0
7.	CESM2-WACCM	USA	320 × 384	POP2	WACCM6
8.	CIESM	China	720 × 560	CIESM-OM	CIESM-AM
9.	CMCC-CM2-SR5	Italy	362 × 292	NEMO3.6	CAM5.3
10.	CMCC-ESM2	Italy	362 × 292	NEMO3.6	CAM5.3
11.	EC-Earth3	Europe	362 × 292	NEMO3.6	IFS cy36r4
12.	EC-Earth3-Veg	Europe	362 × 292	NEMO3.6	IFS cy36r4
13.	EC-Earth3-Veg-LR	Europe	362 × 292	NEMO3.6	IFS cy36r4
14.	FGOALS-g3	China	360 × 218	LICOM3.0	GAMIL3
15.	FIO-ESM-2-0	China	320 × 384	POP2-W	CAM4
16.	GFDL-CM4	USA	1440 × 1080	GFDL-OM4p25	GFDL-AM4.0.1
17.	GFDL-ESM4	USA	720 × 576	GFDL-OM4p5	GFDL-AM4.1
18.	IPSL-CM6A-LR	France	362 × 332	NEMO-OPA	LMDZ
19.	MIROC6	Japan	360 × 256	COCO4.9	CCSR AGCM
20.	MPI-ESM1-2-HR	Germany	802 × 404	MPIOM1.63	ECHAM6.3
21.	MPI-ESM1-2-LR	Germany	256 × 220	MPIOM1.63	ECHAM6.3
22.	MRI-ESM2-0	Japan	360 × 364	MRI.COM4.4	MRI-AGCM3.5
23.	NESM3	China	384 × 362	NEMO v3.4	ECHAM v6.3
24.	NorESM2-LM	Norway	360 × 384	MICOM	CAM-OSLO
25.	NorESM2-MM	Norway	360 × 384	MICOM	CAM-OSLO
26.	TaiESM1	Taiwan	320 × 384	POP2	TaiAM1

amalgamates model data with observations from all over the world to form a globally complete and consistent dataset that adheres to the principles of physics. The ORAS5 relies on the Nucleus for European Modelling of the Ocean (NEMO) ocean model and the NEMOVAR ocean assimilation system. NEMOVAR utilizes the 3D-Var FGAT (First Guess at Appropriate Time) technique to assimilate sub-surface temperature, salinity, sea-ice concentration, and sea-level anomalies.

The ORAS5 dataset is forced by either global atmospheric reanalysis (for the consolidated product) or the ECMWF/IFS operational analysis (for the operational product). The consolidated product (denoted as “Consolidated” in the download form) utilizes the reanalysis atmospheric forcing (ERA-40 until 1978 and ERA-Interim from 1979 to 2014) and re-processed observations. This product is accessible on monthly scales from 1958 to the present at: <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-oras5?tab=overview>. The near real-time (referred to as “Operational” in the download form) ORAS5 product is available from 2015 onwards and is updated on a monthly basis, 15 days behind real-time. It employs ECMWF operational atmospheric forcing and near real-time observations.

The consolidated data benefits from the consistency of atmospheric forcing. The operational data benefits from near real-time latency.

ORAS5, which is an upgraded version of ORAS4 (Balmaseda et al. 2013), has undergone numerous improvements in terms of the ocean’s climate state, variability of SST and sea level, model and data assimilation techniques, as well as in source and usage of observation data sets (Zuo et al. 2019). The ocean model resolution has been increased to 0.25° in the horizontal and 75 levels in the vertical, a significant enhancement compared to 1° and 42 layers in ORAS4.

Further, SST data adopted by ORAS5 is the same as that of ERA5 (Zuo et al. 2019). In ERA5, SST is obtained from two external providers. The HadISST2 dataset is used for SST prior to September 2007, while the OSTIA dataset is used after September 2007. The two datasets largely agree on the global mean SST, suggesting that the combined time series should be consistent over time. However, there are differences in small-scale variations, with the OSTIA dataset performing better in resolving tropical instability waves and sub-mesoscale eddies in the mid-latitudes. This is due to the high-resolution features of the OSTIA dataset, which enable a better fit to scatterometer surface wind observations in

the 4D-Var data assimilation. For a better comparison with CMIP6 model outputs, observation data for SSH and SST was interpolated into the same resolution $1^\circ \times 1^\circ$.

In addition, this study also utilizes SSH data from the National Center for Environmental Prediction (NCEP) Global Ocean Data Assimilation System (GODAS), and the satellite altimeter measurements provided by the Copernicus Climate Change Service (C3S) to corroborate the results of ORAS5 dataset. GODAS is an observationally-based data source that is routinely updated in real-time and its data coverage begins from 1980. It is forced by fields from the NCEP Reanalysis 2, with assimilated temperature and salinity profiles from expendable bathythermographs, moored buoys, and Argo floats through the use of 3DVAR scheme (Saha et al. 2006). This dataset is based on a quasi-global configuration of the GFDL MOM.v3. The model domain extends from 75°S to 65°N and has a resolution of $1^\circ \times 1^\circ$ enhanced to $1/3^\circ$ in the N-S direction within 10° of the equator. The model has 40 levels with a 10 m resolution in the upper 200 m, available at: <https://psl.noaa.gov/data/gridded/data.godas.html>.

The production system for the satellite altimeter SSH measurements provided by C3S utilizes a steady two-satellite merged constellation throughout all time steps. One satellite is designated as the reference to maintain the long-term stability of the data record. The second satellite, which varies over the course of the record, is employed to enhance accuracy, capture mesoscale processes and extend coverage to high latitudes. The dataset is available at <https://cds.climate.copernicus.eu/cdsapp#!/dataset/satellite-sea-level-global?tab=overview>, since the year 1993 at a horizontal resolution of $1/4^\circ \times 1/4^\circ$. Daily fields of SSH data are available, which are further converted to monthly and seasonal dataset through the use of Climate Data Operator.

2.2 ENSO Indices

The study uses the Niño3.4 index to understand the impact of ENSO. The Niño3.4 index is defined as the averaged SST anomaly in the region bounded between 5°S – 5°N , 170°W – 120°W . It is calculated using the SST from ORAS5 reanalysis datasets during 1958–2014. Further, the Niño3.4 indices for CMIP6 models are evaluated for the same period using the SST anomalies of the individual model. Before further analysis, each climate index is detrended and normalized by subtracting the mean and dividing by the standard deviation over the reference period. Figure 1 (a) presents the detrended and normalized Niño3.4 indices of multi-model ensemble mean of 26 CMIP6 models (hereafter MMM), observed dataset, and the CMIP6 model spread of the time-series during DJF from 1958 to 2014. MMM is constructed to reduce biases in individual model outputs (Kharin et al.

2001; Knutti et al. 2010). Here, MMM Niño3.4 index is obtained as an arithmetic mean of 26 CMIP6 Niño3.4 indices. The Niño3.4 index obtained from the observed data clearly portrays the occurrence of strong El Niño events (1965/66; 1972/73; 1982/83; 1991/92; 1997/98; 2009/10) and strong La Niña events (1970/71; 1988/89; 1995/96; 1998/99; 2007/08; 2010/11). Overall, the range of CMIP6 model spread of Niño3.4 time series is consistent with the observed time series.

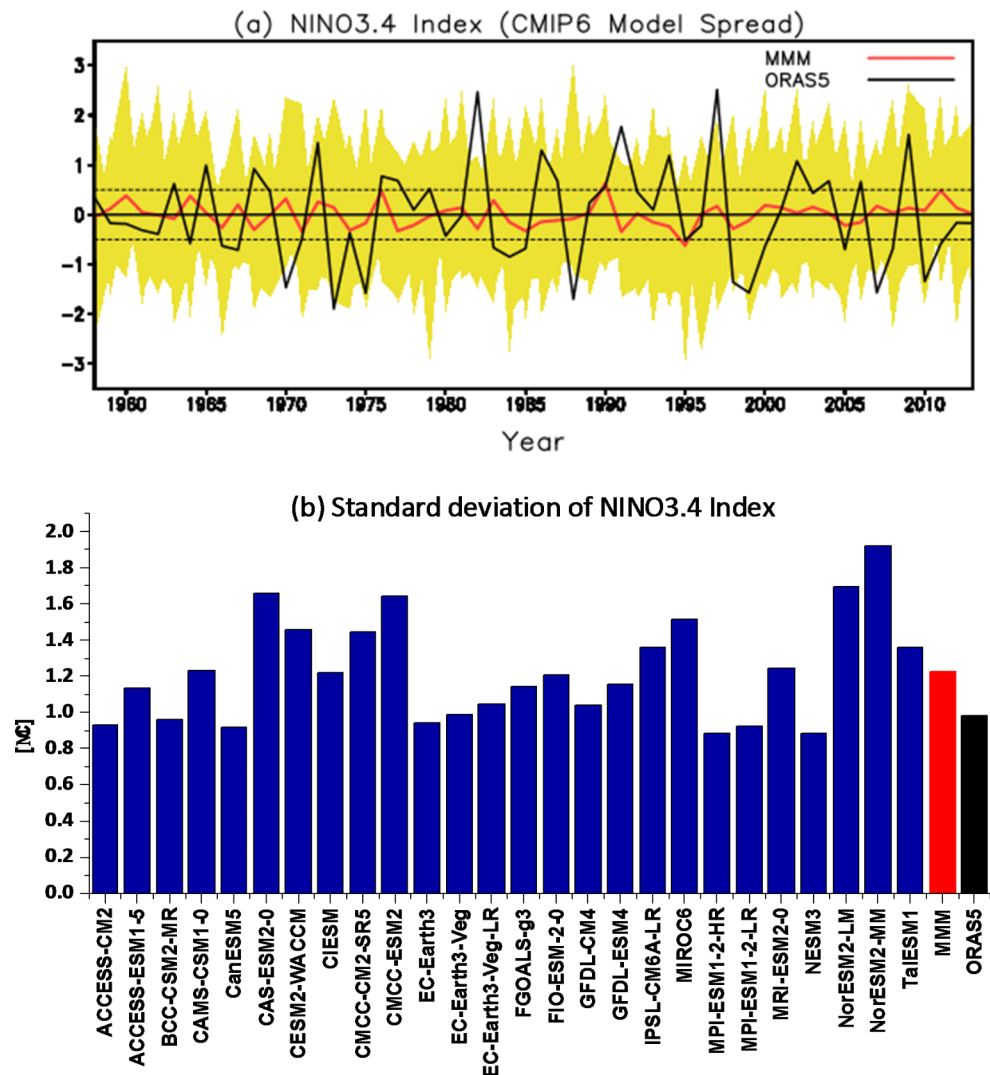
Figure 1 (b) depicts the corresponding interannual standard deviations of Niño3.4 indices before detrending and normalizing. All models exhibit year-to-year fluctuations of the Niño3.4 index, with varying amplitudes across each model. The standard deviation in the Niño3.4 index of the MMM is 1.2°C . Several models (CAM5-CSM1-0, CAS-ESM2-0, CESM2-WACCM, CIESM, CMCC-CM2-SR5, CMCC-ESM2, FIO-ESM-2-0, IPSL-CM6A-LR, MIROC6, MRI-ESM2-0, NorESM2-LM, and NorESM2-MM) exhibit stronger ENSO amplitude ($> 1.25^\circ\text{C}$) than the observed (0.98°C). ACCESS-CM2, BCC-CSM2-MR, EC-Earth3, EC-Earth3-Veg, EC-Earth3-Veg-LR, and MPI-ESM1-2-LR depict similar variabilities to the observation data, yet a few models also simulate weaker ENSO amplitude (MPI-ESM1-2-HR, NESM3).

2.3 Methodology

2.3.1 Bilinear Interpolation

The most commonly used method of climate grid interpolation is bilinear interpolation. It is a simple and efficient method for interpolating data values in two dimensions. This method is easy to use when the source and destination grids are rectilinear. However, curvilinear and unstructured grids may involve sophisticated algorithms to determine the points surrounding the location to be interpolated. Bilinear interpolation is used to estimate the value of a function at a point inside a rectangular grid of known function values, which is implemented in the Climate Data Operator software. The method assumes that the function varies linearly between adjacent grid points in both the x and y directions. Given a rectangular grid of known function values and a point P inside the grid, four nearest grid points to P are determined. These four points define a square in which P lies. The distances between P and each of the four grid points is calculated. These distances are used to weigh the function values at the four grid points. The function at P is interpolated using the weighted average of the function values at the four grid point. The weights are proportional to the inverse of the distances between P and the grid points. For each point that needs to be interpolated, the above steps are repeated.

Fig. 1 (a) Detrended and normalized time series of Niño3.4 indices of CMIP6 MMM represented by red solid line, observation dataset represented by black solid line, and CMIP6 model spread of the time series represented by yellow shading during DJF from 1958–2014, (b) interannual standard deviation of Niño3.4 index (before detrending and normalizing) for CMIP6 models, MMM, and observation data. Horizontal dashed lines in (a) represent ± 0.5 standard deviation



2.3.2 Linear regression

Firstly, the climate indices corresponding to each model, their MMM, and the observation data during DJF are detrended and normalized. The influence of ENSO over SSH/SST is assessed by implementing simple linear regression, wherein the DJF mean of SSH/SST is regressed onto the detrended and normalized ENSO index as:

$$\left. \begin{aligned} SSH &= a_0 + a_1 ENSO \\ SST &= b_0 + b_1 ENSO \end{aligned} \right\} \quad (1)$$

where a_0 , a_1 , b_0 , and b_1 are the regression coefficients, that are estimated using the least square methods. The regression coefficients describe the strength and the type of relationship the explanatory variable has with the dependent variable. Coefficients for positive and negative relationships have positive and negative sign, respectively. Finally, the

spatial distributions of the regression coefficient are displayed as SSH/SST response to ENSO.

Following previous studies (Joshi et al. 2020), the statistical significance of the regression coefficients for each model, their MMM, and observation data is performed via a two-tailed Student's *t*-test. The test statistic is given by $T_r = r\sqrt{n-2}/\sqrt{1-r^2}$ where T_r is the student *t*-value with $(n-2)$ degrees of freedom and r is the correlation coefficient. When $|T_r| > t_{\alpha/2}$ for α level of significance and $t_{\alpha/2}$ as the critical value, it is inferred that there is a significant linear relationship between the variables. The regions with statistical significance passing the α level of significance are shown by hatching.

2.3.3 Skill Metrics

Statistical evaluation of climate models is conducted using various skill metrics such as root mean square error (RMSE), interannual variability skill score (IVS), and M-Score

against the observation SSH response to ENSO during DJF from 1958 to 2014. RMSE is one of the commonly used error index statistics which depicts the differences between the observed and the model data. For each model's SSH response to ENSO, RMSEs are evaluated as follows:

$$RMSE_{XY} = \sqrt{\langle (X - Y)^2 \rangle} \quad (2)$$

where X and Y are denoted as the SSH response to ENSO corresponding to the model and reanalysis dataset, respectively. The angular brackets denote the spatial mean over the global ocean. Better model performance is indicated by a lower RMSE value, and vice versa.

The IVS examines the interannual variability of the simulations relative to that of the observations (Gleckler et al. 2008; Scherrer 2011). It is a symmetric variability statistic used to measure the similarity of interannual variation between the simulation and observation data, which is defined as:

$$IVS = \left(\frac{STD_m}{STD_o} - \frac{STD_o}{STD_m} \right)^2 \quad (3)$$

where STD_m and STD_o denote the spatial standard deviations of the simulation and observation SSH response to ENSO, respectively. A smaller IVS value depicts a better performance of datasets in simulating the interannual variability.

Further, the Arcsin–Mielke measure or the M-Score (Mielke Jr 1991; Watterson 1996; Watterson et al. 2014) represents a non-dimensional skill score based on the MSE normalized by the spatial variance. The M-Score is computed, as defined in Watterson et al. (2015) given by:

$$M = \frac{2}{\pi} \arcsin \left(1 - \frac{MSE}{V_x + V_y + (G_x - G_y)^2} \right) \times 1000 \quad (4)$$

where V_x and V_y represent the spatial variances of the model and observation fields, respectively; G_x and G_y denote the spatial average of the model and observation fields, respectively; MSE is the mean square error between the model field Y and the observation field X; $2/\pi$ is a normalizing factor for the arcsin term which varies from zero to $\pi/2$. The M-Score can range from zero (or even below), indicating no skill, to a theoretical maximum of 1000 when the $MSE = 0$. The findings of the above given three metrics are quantified in Table 2. Based on these skill metrics, individual models are ranked accordingly, and a total rank is obtained by taking an arithmetic average of the above three metric rankings.

Additionally, Taylor Skill Score (TSS) relates the correlation coefficient and standard deviations of the models to observations (Taylor 2001). TSS is an important statistical tool to evaluate climate models, as documented in many studies (Kusunoki and Arakawa 2015). The TSS is estimated as follows:

$$TSS = \frac{4(1 + R)}{\left(\tilde{\sigma}_f + \frac{1}{\tilde{\sigma}_f} \right)^2 (1 + R_o)} \quad \text{with } \tilde{\sigma}_f = \frac{\sigma_f}{\sigma_o} \quad (5)$$

where R represents the correlation coefficient between the model simulated and observation SSH response to ENSO, and $\tilde{\sigma}_f$ is the ratio of the standard deviation of the model simulation to that of observation. R_o is the maximum correlation coefficient attainable and assumed to be 1 for this analysis. A TSS value close to 1 represents a better skill for the model.

Table 2 Root-Mean-Square Errors (RMSEs), Interannual Variability Skill Score (IVS), and M-Score of individual CMIP6 Models against the Observation SSH response to ENSO for DJF during 1958–2014

Model Number	Model Name	RMSE (m)	IVS	M-Score
1.	ACCESS-CM2	0.0111	0.447	583
2.	ACCESS-ESM1-5	0.0125	0.625	516
3.	BCC-CSM2-MR	0.0133	1.138	466
4.	CAMS-CSM1-0	0.0110	0.066	619
5.	CanESM5	0.0129	1.721	470
6.	CAS-ESM2-0	0.0137	0.0004	547
7.	CESM2-WACCM	0.0111	0.002	640
8.	CIESM	0.0105	0.003	663
9.	CMCC-CM2-SR5	0.0123	0.018	612
10.	CMCC-ESM2	0.0119	0.089	634
11.	EC-Earth3	0.0099	0.252	639
12.	EC-Earth3-Veg	0.0095	0.309	651
13.	EC-Earth3-Veg-LR	0.0092	0.011	695
14.	FGOALS-g3	0.0125	0.488	524
15.	FIO-ESM-2-0	0.0104	0.028	645
16.	GFDL-CM4	0.0111	0.280	592
17.	GFDL-ESM4	0.0110	0.280	587
18.	IPSL-CM6A-LR	0.0105	0.203	620
19.	MIROC6	0.0122	0.089	632
20.	MPI-ESM1-2-HR	0.0143	0.916	433
21.	MPI-ESM1-2-LR	0.0138	1.221	444
22.	MRI-ESM2-0	0.0101	0.015	658
23.	NESM3	0.0125	1.721	486
24.	NorESM2-LM	0.0121	0.002	603
25.	NorESM2-MM	0.0150	0.282	578
26.	TaiESM1	0.0138	0.174	597
27.	MMM	0.0085	0.092	693

3 Results

3.1 Climatology and variability

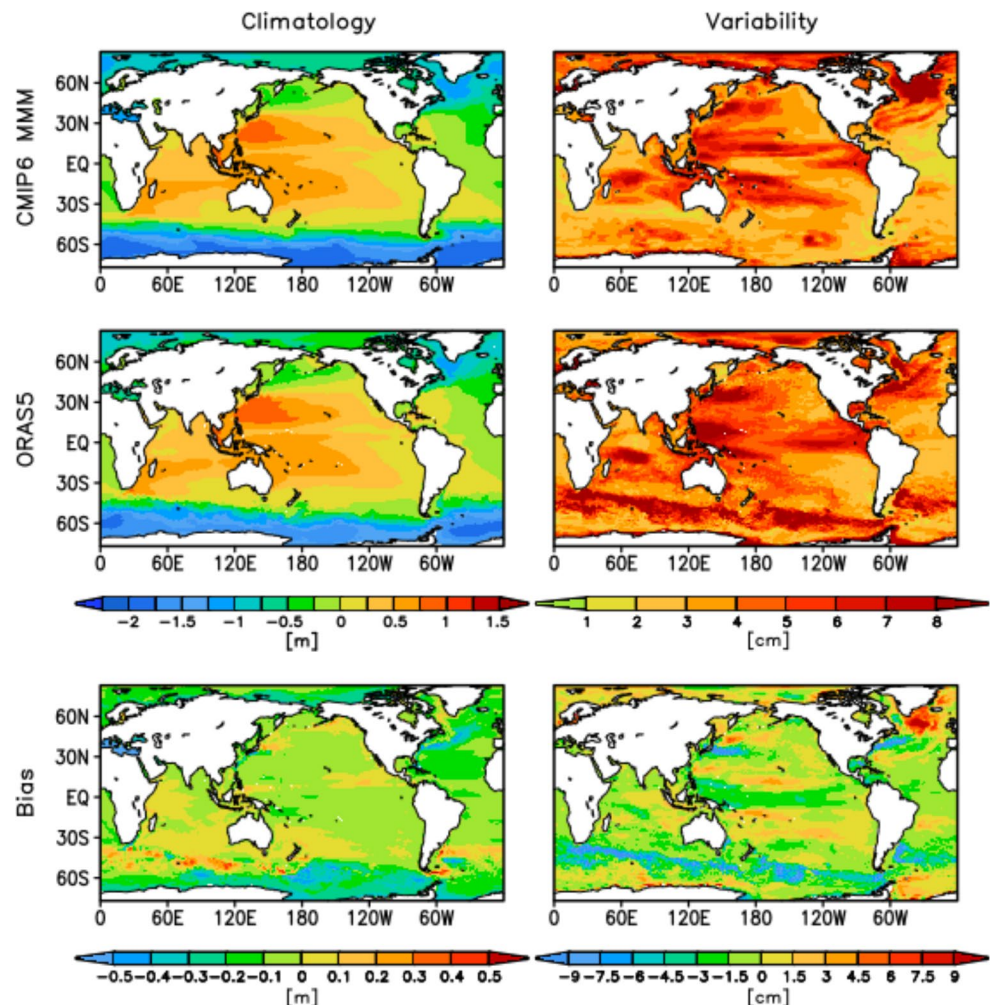
The spatial climatology and variability pattern of SSH during DJF averaged over the period 1958–2014 is presented in Fig. 2 for CMIP6 MMM and ORAS5 (observation data). Here, MMM corresponds to a simple arithmetic average across individual models. MMM serves as a valuable tool in climate model evaluation by reducing model biases, quantifying uncertainty, exploring model spread and consensus, providing policy-relevant information through a robust synthesis of climate model results and gaining comprehensive insights into potential climate outcomes. In addition, bias in the climatology and variability of CMIP6 MMM (i.e., MMM minus ORAS5) is provided in the last row of Fig. 2.

The climatology pattern of observation data exhibits positive SSH in the subtropical gyres in both the hemispheres north and south of the Equator. On the other hand, negative SSH is evident in the subpolar gyres around 50°S in the PO (moderate) and 60°S in the AO (strong). The transition from yellow to green over the Southern Ocean (SO) in the

climatological patterns represents the northern boundary of the Antarctic Circumpolar Current (ACC), also referred to as the northern edge of the Subantarctic Front (SAF), consistent with the study of Dietze et al. (2020). Along the KE, the westward propagating Rossby waves cause varying effects on the coastal SSH of Japan. Due to the meridional shifts of the Kuroshio Path, the excited coastal waves propagate and induce a large SSH signal along the coast of Japan (Fedele et al. 2021). Overall, the climatology of CMIP6 MMM and observation data yield a good spatial correlation ($r=0.96$). Thus, CMIP6 MMM reproduces the pattern well and appropriately captures the main features of the global barotropic circulation. Yet, CMIP6 MMM underestimates SSH over large parts of PO and AO by 10 cm. Also, a slight overestimation is noted in the northwest IO and southeastern sector of southern IO by 10 cm.

Further, the variability in SSH is represented by the inter-annual standard deviation. ORAS5 SSH shows strong inter-annual variability over the western tropical PO and eastern tropical PO, with an east-west dipole pattern over the tropical PO during DJF, consistent with the previous studies (Cazenave and Nerem 2004). Additionally, strong SSH

Fig. 2 Climatology and Variability of SSH for CMIP6 MMM and ORAS5 data during DJF from 1958–2014. Note that ‘variability’ is the statistic interannual standard deviation. Further, the bias (MMM–ORAS5) in SSH mean and standard deviation is shown in the last row



variability occurs near the continental margins of northwest PO and AO, western sector of tropical IO, and SO where active ocean currents prevail. Yet, low interannual variability is evident in the tropical AO and southeast PO. Changes in ocean currents play an essential role in the interannual variability of SSH, and large SSH signals over the subtropics may be associated with ocean processes (Chen et al. 2023). Overall, CMIP6 MMM reproduces the SSH variability well and exhibits similar patterns to ORAS5 SSH variability. Minimal biases (2–3 cm) prevail in depicting the regional SSH variability except for the north AO, where overestimation reaches up to 9 cm, and the SO, where the underestimation reaches up to 9 cm. Also, the overall patterns of SSH variability in ORAS5 (Fig. 2) resemble those from GODAS and satellite altimeter datasets (Fig. S1). Good agreement of SSH variability between satellite altimeter and reanalysis products indicates the robustness of our results.

3.2 Model evaluation

3.2.1 SSH and SST response to ENSO

The spatial pattern of regression coefficients of SSH onto Niño3.4 index for CMIP6 models, their MMM, and ORAS5 dataset during DJF is depicted in Fig. 3. Hatching indicates regions with statistical significance at the 5% level. For MMM, an ensemble mean of all the regression coefficients is displayed. By combining the outputs of multiple models through a MMM, it is possible to reduce the impact of individual biases or errors, and obtain a more robust estimate. Based on the ORAS5, El Niño events instigate significant enhancement in SSH over the eastern equatorial PO and southwest tropical IO, while a significant reduction in SSH is apparent over the region bounded between 100°E–133°E, 5°S–20°N (Region II), which extends towards the western tropical PO. Overall, SSH response to ENSO across global oceans is moderately captured by various individual models. However, some models (CESM2-WACCM, CMCC-ESM2, CMCC-CM2-SR5, FIO-ESM2-0, MIROC6, NorESM2-MM, NorESM2-LM, and TaiESM1) overestimate the observed pattern. While some (CanESM5, GFDL-CM4, GFDL-ESM4, ACCESS-ESM1-5, MPI-ESM1-2-LR, MPI-ESM1-1-HR, and NESM3) underestimate the observed pattern. CIESM, MRI-ESM2-0, and EC-Earth3-Veg-LR demonstrate a similar SSH response pattern to the observation data. Also, GODAS and satellite altimeter dataset align with ORAS5 in illustrating the SSH response to ENSO (Fig. S1), thereby highlighting the reliability of our findings.

The correlation coefficients between the CMIP6 models and the observation SSH regression pattern range from 0.71 to 0.92. The MMM regression pattern of SSH exhibits its maximum correlation (0.92) with that of the observation

pattern. Despite high agreement between the regression pattern of CMIP6 models and ORAS5, bias maps portray some regional differences in magnitude. The bias across models (Model–ORAS5) in the representation of SSH regression onto Niño3.4 index is displayed in Fig. S2. Models such as ACCESS-ESM1-5, CanESM5, CAS-ESM2-0, MPI-ESM1-2-HR, MPI-ESM1-2-LR, and NESM3 exhibit a positive bias in the regression pattern over the western PO. Some models (NorESM2-MM, TaiESM1) also exhibit positive bias over the southwest tropical IO. Yet, negative biases are evident across models over the eastern PO (ACCESS-CM2, ACCESS-ESM1-5, BCC-CSM2-MR, CanESM5, FGOALS-g3, MPI-ESM1-2-HR, MPI-ESM1-2-LR, and NESM3). Overall, MMM performs better than individual models and exhibits minimal biases in the representation of SSH regression pattern.

In Fig. 4, the spatial patterns of regression coefficients of SST onto Niño3.4 for CMIP6 models, MMM, and observation dataset are shown. Warmer SST signals prevail over the southwest tropical IO and eastern equatorial PO. On the other hand, cooler conditions prevail over the western tropical PO in response to ENSO. Although most of the models appropriately capture the SST response, some models (CAS-ESM2-0, CMCC-ESM2, CMCC-CM2-SR5, IPSL-CM6A-LR, MIROC6, and NorESM2-MM) exhibit a westward shift of eastern PO signal and overestimate the observed pattern. While ACCESS-CM2, BCC-CSM2-MR, CAMS-CSM1-0, EC-Earth3, MPI-ESM1-LR, and MPI-ESM1-HR models underestimate the observed patterns. For Region II, ACCESS-ESM1-5, CanESM5, and IPSL-CM6A-LR exhibit almost no response. Overall, the observed SST response to ENSO remarkably resembles the SSH response to ENSO, indicative of a robust teleconnection at work between SSH and SST.

In Fig. 5, estimates from the Taylor diagram analysis quantify the model skills for the SSH and SST response to ENSO. Overall, skills for SSH response exhibit a good correlation varying between 0.75 and 0.90. Likewise, for the SST response, the correlation varies between 0.79 and 0.90. Overall, both SSH and SST responses exhibit comparable skills. Moreover, due to bias cancellation, MMM performs better than many individual models for both SSH and SST responses. Some models with comparatively higher standard deviation in SSH response are well identified (EC-Earth3-Veg, MIROC6, TaiESM1, and NorESM2-MM). Models that are located in the vicinity of the observation reference point (or perfect model) can be considered as better skilled models (Fig. 5a), which include CIESM, CAS-ESM2-0, NorESM2-LM, CESM2-WACCM, CMCC-CM2-SR5, GFDL-CM4, CMCC-ESM2, NESM3, and FIO-ESM2-0. The resulting skill score confirms these evaluation results.

Fig. 3 Spatial distribution of the regression coefficients of SSH onto the Niño3.4 index during DJF from 1958–2014 for CMIP6 models, MMM, and ORAS5 data. Each panel of the figure contains the correlation coefficient of SSH regression pattern of model with that of observation data in brackets. Hatching represents statistically significant regions at the 5% level. The unit of regression coefficients is m per standard deviation

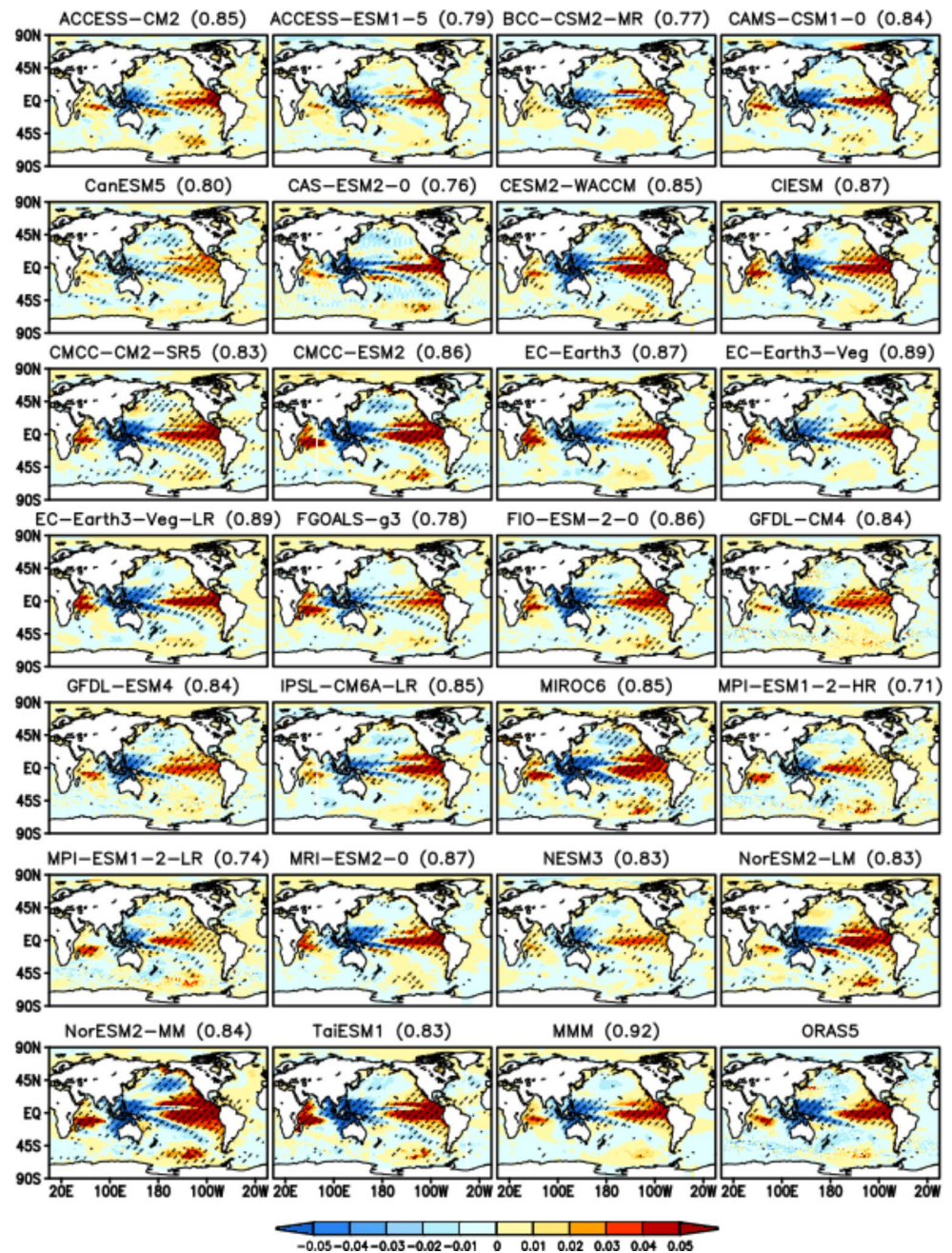
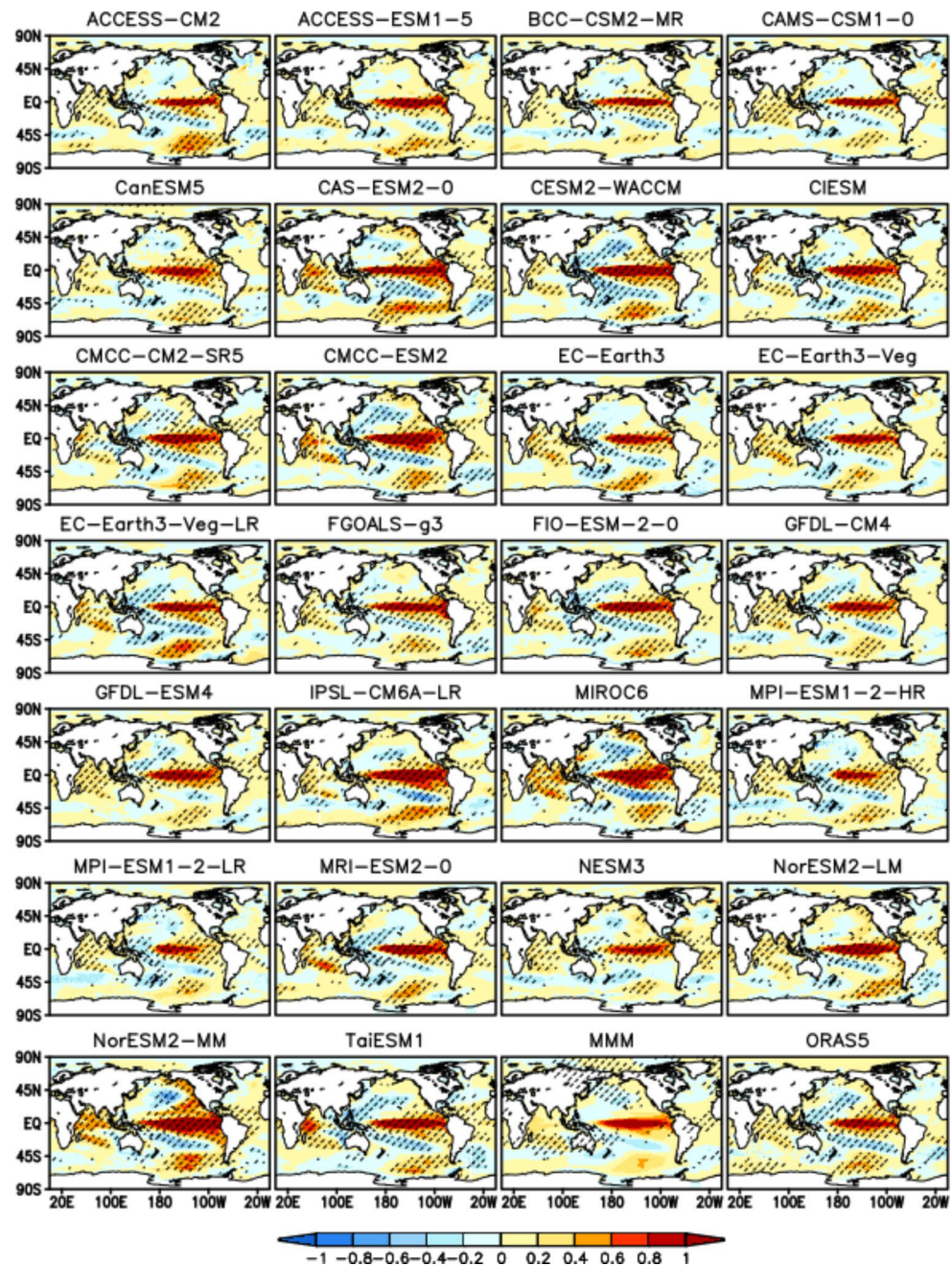


Figure S3 depicts the inter-model spread in the ENSO SSH metrics used in Fig. 5. Inter-model spread refers to the variability among different climate models in their outputs for a specific metric. It is clearly evident that the correlation coefficient exhibits the least inter-model spread, followed by the Taylor skill Score. This indicates a remarkable consensus among the models in capturing the overall relationship between ENSO and SSH, thereby highlighting the robustness of climate models. On the other hand, standard deviation exhibits large inter-model spread, which also underscores the potential limitations in certain aspects of the model's simulation, such as model structures, initial

conditions, parameterizations, and other factors that influence their behaviour.

For SST response to ENSO, models identified with relatively larger standard deviation are CAS-ESM2-0, CMCC-ESM2, NorESM2-MM, IPSL-CM6A-LR, MIROC6, and TaiESM1. Some models, such as CAMS-CSM1-0, EC-Earth3-Veg-LR, FIO-ESM2-0, and GFDL-ESM4, are closer to the observation reference point (see Fig. 5b), and may be considered as more skilled in depicting SST response to ENSO than others.

Fig. 4 Same as Fig. 3, but for SST. The unit of regression coefficient is $^{\circ}\text{C}$ per standard deviation



3.2.2 Model evaluation based on skill metrics

In this section, CMIP6 model performance for SSH response is assessed using three skill metrics as RMSE, IVS, and M-Score. Based on skill metric parameters, each model is ranked according to their performance (as shown in Table 2). Rankings are graphically represented using a portrait diagram in Fig. 6, which offers a concise summary of the CMIP6 global historical simulation of model SSH response compared to the observed SSH response. For an overall evaluation, a total ranking is also determined from

the arithmetic average of the three rankings (last column of Fig. 6).

Among all the CMIP6 models, EC-Earth3-Veg-LR exhibits the least RMSE value, followed by EC-Earth3-Veg, EC-Earth3, MRI-ESM2-0, FIO-ESM2-0, CIESM, IPSL-CM6A-LR, CAMS-CSM1-0, GFDL-ESM4, and ACCESS-CM2. Overall, the RMSE values of the model simulations relative to the observation SSH responses are minimal, thereby indicating that most of the CMIP6 models appropriately capture the SSH response to ENSO during boreal winters. The MMM exhibits the lowest RMSE value

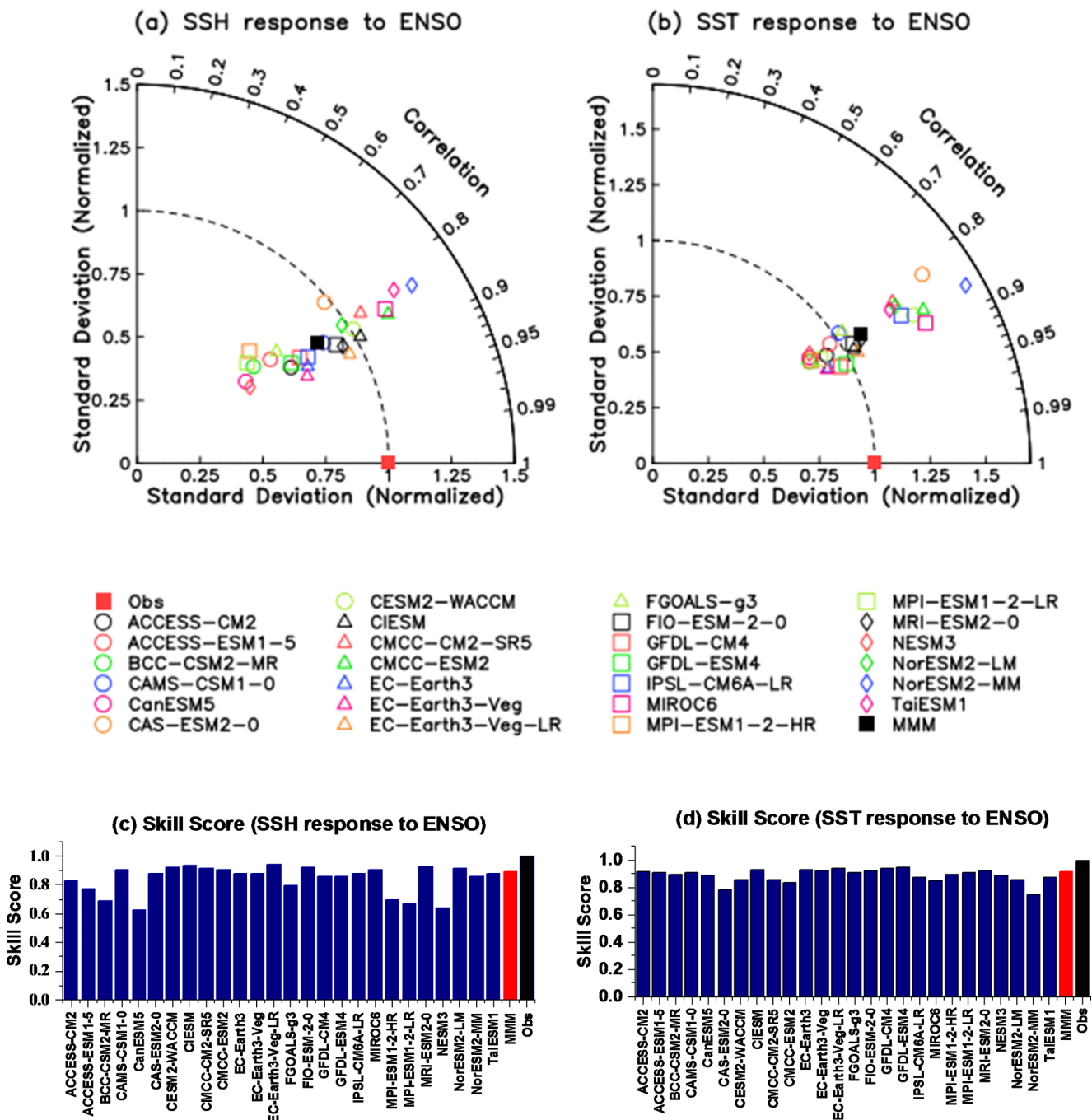


Fig. 5 Taylor diagram for (a) SSH and (b) SST response to Niño3.4 for CMIP6 models, MMM, and observation data. The standard deviation of each modeled variable is normalized to the standard deviation of the observations. A perfect simulation would lie at 1 on the abscissa repre-

compared to CMIP6 models, and thus outperforms the individual models as expected.

IVS values are calculated for each model based on SSH response to ENSO over the global ocean (see Table 2). Among all the CMIP6 models, CAS-ESM2-0, CESM2-WACCM, NorESM2-LM, CIESM, EC-Earth3-Veg-LR, CMCC-CM2-SR5, MRI-ESM2-0, FIO-ESM2-0, CAMS-CSM1-0, and

sented by the observation data. The Taylor skill scores for (c) SSH and (d) SST response to Niño3.4 for CMIP6 models, MMM, and observation data is given

MIROC6 are ranked as top ten models in the evaluation of IVS for SSH response. The top ten models in evaluating IVS for SSH response exhibit lower value than MMM IVS value. Models such as CanESM5 and NESM3 show high IVS values, thus poorly simulate the interannual variability of winter SSH response to ENSO.

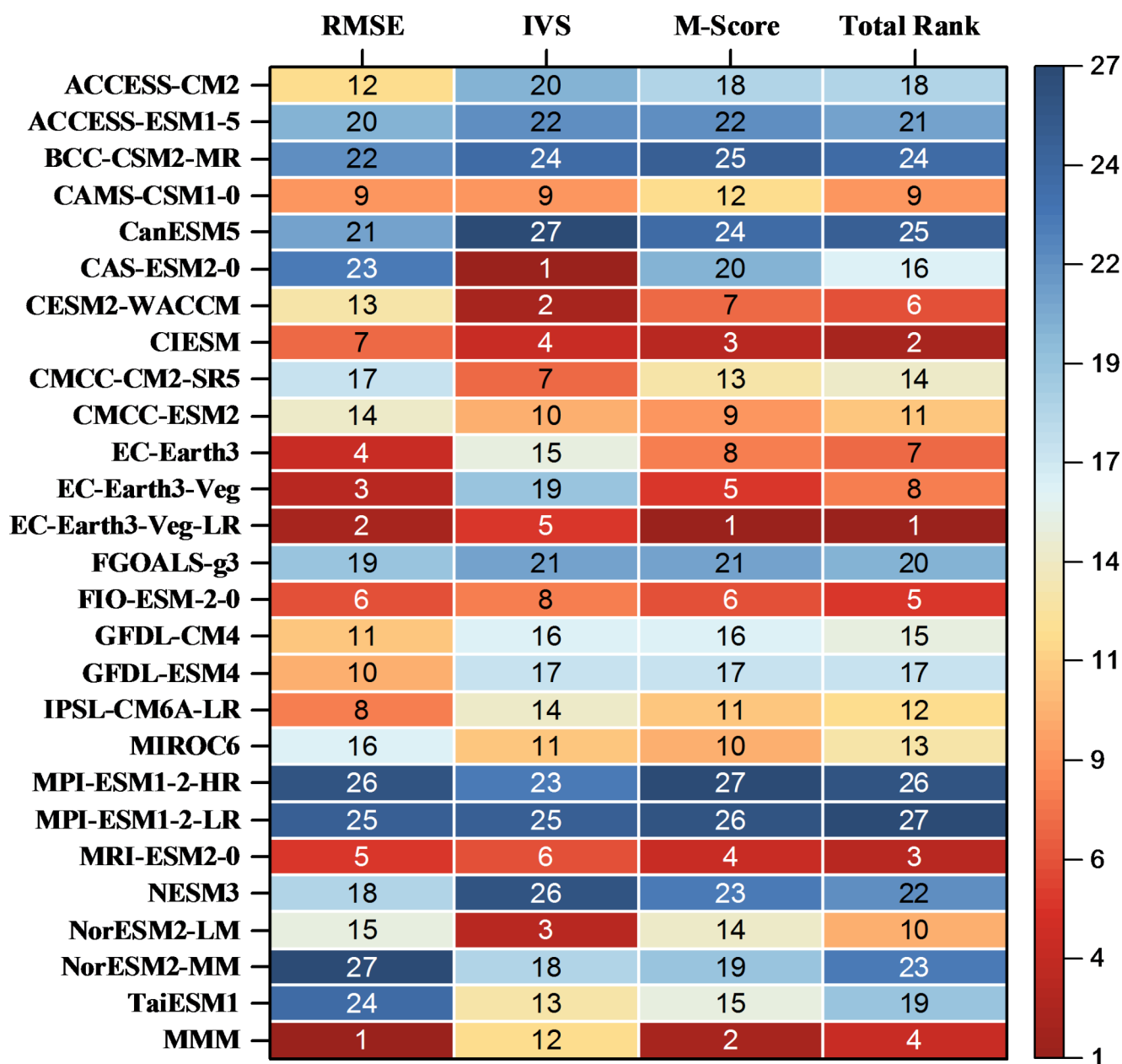


Fig. 6 Portrait diagram is shown for the rankings based on RMSE, IVS, and M-Score of individual CMIP6 model against the observation SSH response to ENSO based on the evaluations made in Table 2 dur-

ing DJF from 1958–2014. Total rank is obtained by taking an arithmetic average of the three metric rankings

Further, M-Score evaluated for the SSH response to ENSO among the CMIP6 models range between 400 and 700. A higher value of M-Score corresponds to a better performance of the CMIP6 models. EC-Earth3-Veg-LR, MMM, CIESM, MRI-ESM2-0, EC-Earth3-Veg, FIO-ESM2-0, CESM2-WACCM, EC-Earth3, CMCC-ESM2, and MIROC6 models are top ten models for the evaluation of M-Score. The lowest M-Score prevails for BCC-CSM2-MR, CanESM5, MPI-ESM1-2-HR, and MPI-ESM1-2-LR.

Overall, it is observed that the performance of CMIP6 models varies differently in individual skill metric parameters

(RMSE, IVS, and M-Score). Specifically, CMIP6 models exhibit a range of performances in the representation of SSH response to ENSO based on different skill metric parameters. Besides, ranking models' performance based on one single parameter would not be sufficient to give a detailed outcome. Therefore, an arithmetic average of the rank of these skill metrics has been considered to determine the total rank of individual models, which will provide a synthesis of the SSH response to ENSO for each model. Based on the total ranking, the top best-performing models are EC-Earth3-Veg-LR, CIESM, MRI-ESM2-0, MMM,

FIO-ESM2-0, CESM2-WACCM, EC-Earth3, EC-Earth3-Veg, CAMS-CSM1-0, and NorESM2-LM (see Fig. 6).

3.3 Model evaluation of the hotspot regions

Based on the observation response strength of SSH and SST regressed onto the ENSO index, three hotspot regions are selected as eastern PO referred to as Region I (170° W–95° W, 10° S–7° N), Region II (100° E–133° E, 5° S–20° N), and southwest tropical IO referred to as Region III (47° E–84° E, 17° S–3° S) to examine which factors determine model skills on SSH and SST responses to ENSO. Significant enhancement in SSH is evident over the eastern PO and southwestern tropical PO, while reduction over Region II. Numerous studies support that these areas exhibit dominant ENSO influences (Han et al. 2017; Hamlington et al. 2019). The selected regions are influenced by broader considerations, including their socioeconomic factors, given their particular importance and vulnerabilities related to ENSO impacts on SSH. Evaluating climate model accuracy in simulating these impacts can have significant implications for the well-being of communities, industries, and ecosystems along the coastline of these regions. For each hotspot region, the relationship between area mean regression coefficients for SSH, SST, and ENSO amplitude is studied using an inter-model correlation analysis based on CMIP6 models.

For the eastern PO (Fig. 7), moderate inter-model correlation ($r=0.77$, significant at the 5% level) between ENSO amplitude and SSH regression onto ENSO index reveals that ENSO is a driving factor for prominent variations in the inter-model SSH regression coefficients over this region. Indeed, models with stronger ENSO amplitude tend to simulate stronger regression of SSH with ENSO (TaiESM1, CMCC-CM2-SR5, MIROC6, CAS-ESM2-0, CMCC-ESM2, NorESM2-MM, and NorESM2-MM) and vice versa (BCC-CSM2-MR, CanESM5, NESM3, MPI-ESM1-2-LR, and MPI-ESM1-2-HR). Large model spread is evident over the eastern PO. Likewise, a strong correlation ($r=0.97$, significant at the 1% level) is obtained between the ENSO amplitude and regression of SST with ENSO. Models with stronger ENSO amplitude are NorESM2-MM, NorESM2-LM, CAS-ESM2-0, CMCC-ESM2, MIROC6, and CMCC-CM2-SR5. While models with weaker ENSO amplitude and SST regression with ENSO are MPI-ESM1-2-HR, NESM3, MPI-ESM1-2-LR, and BCC-CSM2-MR. Further, a good correlation ($r=0.77$, significant at the 5% level) is estimated between the area mean regression coefficient for SSH and SST over the eastern PO. This indicates that oceanic warming plays a vital role to enhance the SSH regression coefficient over the eastern PO.

In Region II (Fig. 8), a negative correlation of 0.63 (significant at the 5% level) is obtained between the ENSO

amplitude and SSH regression coefficient. Precisely, models with stronger ENSO amplitude (CMCC-ESM2, NorESM2-LM, NorESM2-MM, CAS-ESM2-0, MIROC6, and CESM2-WACCM) are characterized by strengthened negative correlation between SSH regression and ENSO amplitude. While models with weaker ENSO amplitude are BCC-CSM2-MR, EC-Earth3, NESM3, MPI-ESM1-2-HR, and MPI-ESM1-2-LR. Likewise, ENSO amplitude is found to have a negative correlation with SST regression (-0.30). Further, a moderate correlation ($r=0.65$, significant at the 5% level) is obtained between the area mean regression coefficient for SSH and SST over Region II. This reflects that the SST teleconnection patterns are fundamental to simulate SSH regression onto ENSO over this region. In particular, ENSO induces significant changes in SST, which in turn leads to reduced SSH over this region, whereby upwelling Kelvin waves peak along the Sumatra/Java coasts (Delman et al. 2016).

Further, moderate inter-model correlation ($r=0.52$, significant at the 5% level) is obtained between the ENSO amplitude and SSH regression over the southwest tropical IO (Fig. 9). Similarly, ENSO amplitude exhibits a moderate positive relationship ($r=0.43$, significant at the 5% level) with SST regression over this region. Regression of SST is stronger in models such as NorESM2-MM, NorESM2-LM, CAS-ESM2-0, CMCC-ESM2, and MIROC6. While weaker regression of SST is simulated in models such as ACCESS-CM2, BCC-CSM2-MR, CanESM5, EC-Earth3, MPI-ESM1-2-LR, MPI-ESM1-2-HR, and NESM3. Overall, SSH and SST regression to ENSO exhibits a moderate correlation ($r=0.60$, significant at the 5% level) over this region. Enhanced SSH signals over this region depict westward propagating equatorial Rossby waves that are formed due to strong wind forcing or through the reflection of Kelvin waves at the eastern boundary (Phillips HE et al. 2021). The westward propagating Rossby waves partly generated by the El Niño-induced anticyclonic wind stress curl in the eastern basin strongly associate with the SST variability in the southwestern TIO (Chowdary et al. 2009; Gnanaseelan and Vaid 2010).

Figure S4 represents the model spread in the area mean regression coefficient of SSH and SST over three hotspot regions for individual models (unshaded circles), ensemble mean of top ten models (red solid circle), top twenty models (green solid circle), and the MMM (bar graph). Model spread for SSH regression coefficient is highest over Region III, while it is lowest for Regions I and II. Contrarily, model spread for SST area mean regression coefficient is greatest in region I but least pronounced over Regions II and III. Ensemble mean for top ten and top twenty models, and the MMM demonstrates relatively minor differences in the extent of spread.

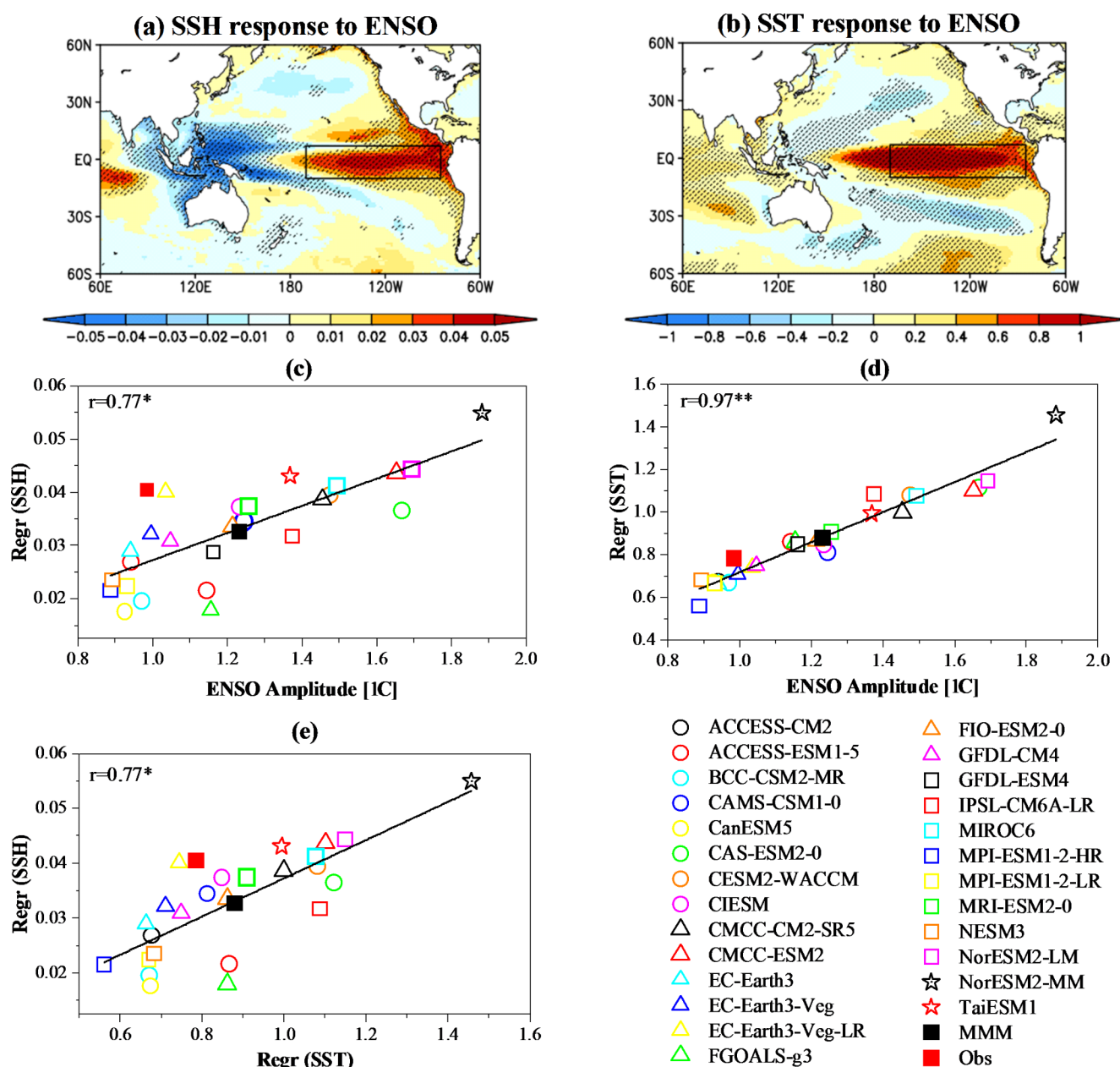


Fig. 7 (a) Same as MMM of Fig. 3 (b) Same as MMM of Fig. 4 but for Region I (eastern PO; 170° W–95° W, 10° S–7° N), (c)–(e) scatter plots and linear regression lines (solid black lines) representing the relationship between (c) ENSO amplitude and SSH regression with

ENSO (d) ENSO amplitude and SST regression with ENSO (e) regression of SSH and SST with ENSO across CMIP6 models. Inter-model correlation coefficient (r) is mentioned. * and ** indicate the statistically significant correlation at the 5% and 1% level, respectively

4 Conclusion and discussion

The study assesses 26 CMIP6 models' ability to simulate SSH response to ENSO during DJF across global oceans. SSH and SST responses to ENSO are analyzed by using simple linear regression analysis. During the positive phase of ENSO, enhanced SSH appears over the eastern PO (Region I) and southwest tropical IO (Region III), while reduced SSH occurs over the region bounded between 100° E–133° E, 5°

S–20° N (Region II), attributed to relaxed trade winds over the tropical PO (and Church et al. 2013; Han et al. 2017; Lyu et al. 2017; Kumar et al. 2022). SST responses, which mirror SSH patterns, are propelled more by local wind circulation changes. Furthermore, model evaluation based on SSH regression pattern using different skill metric analysis (RMSE, IVS, and M-Score) identifies the best-performing models, with the following in the highest-ranked positions: EC-Earth3-Veg-LR, CIESM, MRI-ESM2-0, MMM, FIO-ESM2-0, CESM2-WACCM, EC-Earth3, EC-Earth3-Veg,

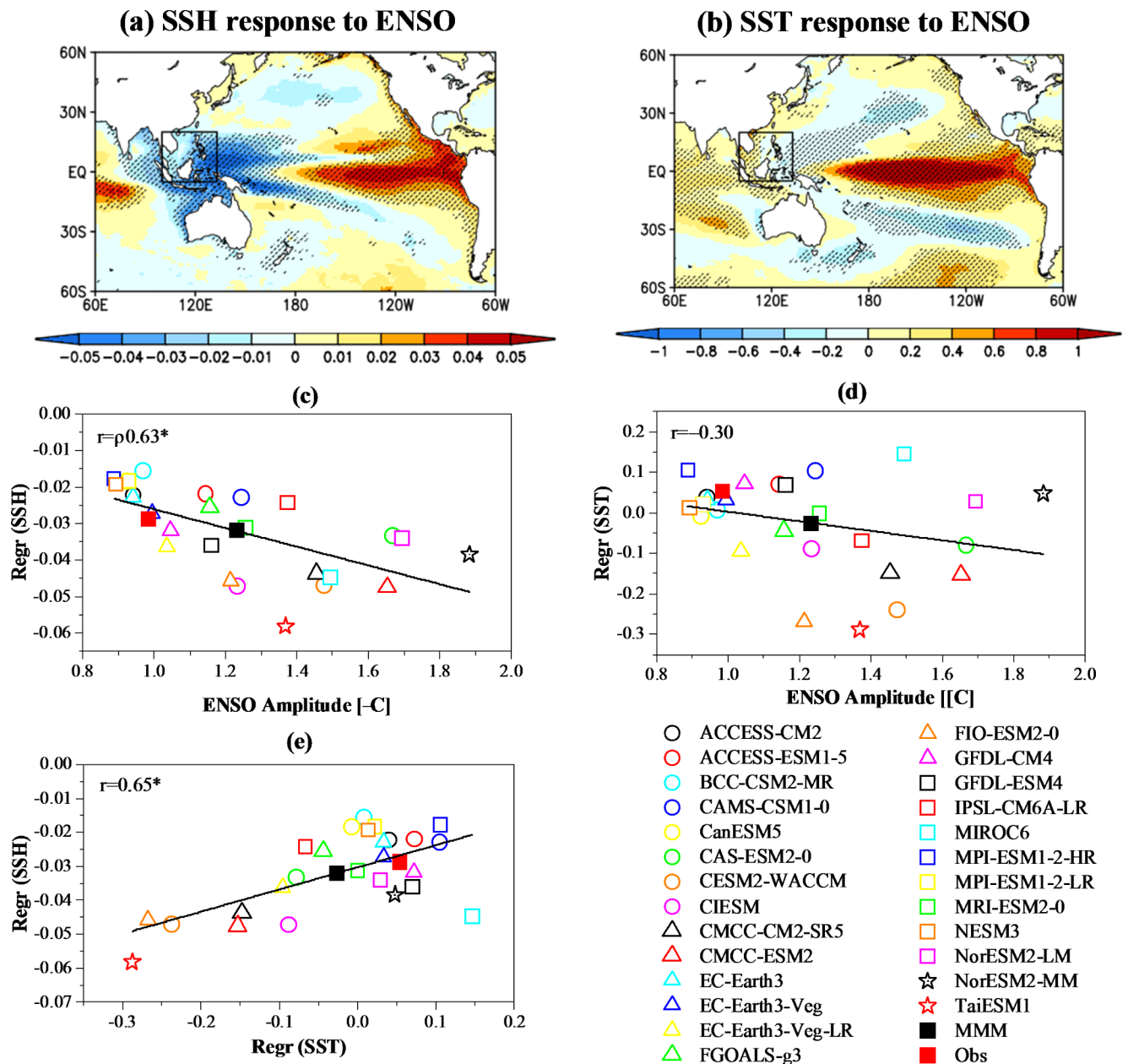


Fig. 8 Same as Fig. 7 but for Region II (100° E–133° E, 5° S–20° N)

CAMS-CSM1-0, and NorESM2-LM. The inter-model relationship between ENSO characteristics and area mean regression coefficient of SSH/SST reveals strengthened positive correlation over Regions I and III while negative correlation over Region III. Overall, the salient contribution of this study lies in its noteworthy implication that the inclusion of ENSO intensity and atmospheric teleconnection patterns is imperative for achieving better simulations and reliable predictions of SSH, thereby advancing our understanding of the dynamics of the oceanic systems.

Limitations of this study involve diverse resolutions among CMIP6 climate models, thereby affecting their

ability to capture small-scale features like eddies. Enhancing the fidelity of eddy representation within the climate models holds significance due to their crucial role in redistributing heat, nutrients, various oceanic properties, influencing large-scale climate patterns and interactions between ocean and atmosphere. Another constraint of this study is its sole focus on the DJF season, characterized by the peak intensity of ENSO. Consequently, its impact on SSH holds greater prominence compared to other seasons, aligning with the findings from previous research (Kumar et al. 2022). Given this, further investigation is warranted to understand the detailed physical processes of how different

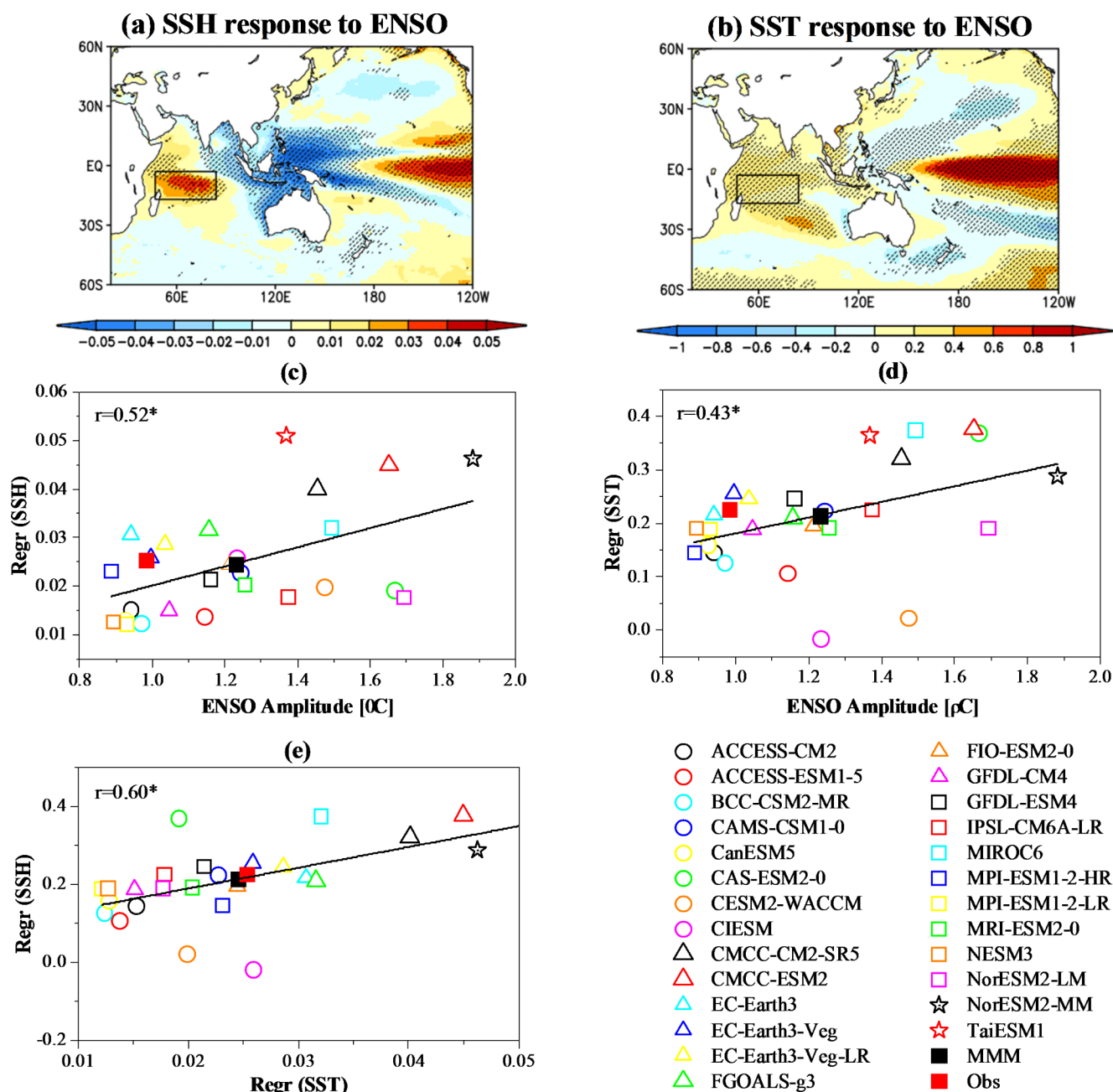


Fig. 9 Same as Fig. 7 but for Region III (southwest tropical IO; 47° E–84° E, 17° S–3° S)

climate modes interact with ENSO across different seasons and induce varying SSH changes. Moreover, considering factors like ocean currents, wind patterns, sea ice, topography, bathymetry, and anthropogenic factors could provide a better understanding of SSH variability.

Further, the study opens avenues to explore how the identified regions and their SSH responses to ENSO align with historical and future climate change scenarios. Understanding the evolution of these responses under various greenhouse gas emission scenarios will aid to a broader

comprehension of the implications of ENSO-driven SSH patterns within the context of changing climate dynamics.

List of abbreviations

ENSO	El Niño Southern Oscillation
DJF	December–February
SSH	Sea surface height
MMM	Multi-model ensemble Mean
CMIP6	Coupled Model Intercomparison Project Phase 6
SST	Sea surface temperature

PO	Pacific Ocean
Kuroshio Extension	KE
Indian Ocean	IO
MC	Maritime Continent
GCMs	Global climate models
ESGF	Earth System Grid Federation
ORAS5	Ocean Reanalysis System 5
ECMWF	European Centre for Medium-Range Weather Forecasts
NEMO	Nucleus for European Modelling of the Ocean
NCEP	National Center for Environmental Prediction
GODAS	Global Ocean Data Assimilation System
C3S	Copernicus Climate Change Service
RMSE	Root mean square error
IVS	Interannual variability skill score
TSS	Taylor skill score
ACC	Antarctic Circumpolar Current
SAF	Subantarctic Front
AO	Atlantic Ocean

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Authors' contributions Divya Sardana: Data curation, Investigation, Writing- Original draft preparation, Validation; Prashant Kumar: Conceptualization, Visualization, Methodology, Supervision, Writing-Reviewing and Editing; Rajni: Visualization, Writing- Reviewing and Editing;

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Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not Applicable.

Consent for publication I, the undersigned, give my consent for the publication of identifiable details, which can include photograph(s) and/or videos and/or case history and/or details within the text ("Material") to be published in the above Journal and Article.

Competing interests The authors declare that they have no known competing financial interests.

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